

# List of Problems about Knowledge—Teaching remarks for selected problems

ECO403/503—Introduction to Contract Theory and Auctions

Alejandro Melo Ponce

Nazarbayev University

1. I love Juliet, and I know that Juliet loves me, but I do not know if Juliet knows that I love her. Use a knowledge space this situation, and specify the state of the world in that knowledge space that corresponds to the situation described.

## Solution

- $N = \{I, (J)\text{uliet}\}$ .
- When solving this problem, most people will be tempted to use four states only (corresponding to each possible configuration of love for  $I$ , and  $J$ ). However, this wouldn't be correct because such a state space wouldn't allow us to discuss the **second order level** of knowledge that the problem is asking for: I don't know if Juliet knows that I love her. Thus, we need five states:
  - $\omega_1$  and  $\omega_2$  corresponding to the situation in which I Love Juliet and Juliet loves me; in the first one Juliet knows that I love her, but in the second one she does not.
  - $\omega_3$  is the situation in which I love Juliet but she does not love me.
  - $\omega_4$  is the situation in which I do not love Juliet but Juliet loves me.
  - $\omega_5$  is the situation in which I do not love Juliet and Juliet does not love me.
- The information partitions:  $\{\{\omega_1, \omega_2\}, \{\omega_3\}, \{\omega_4, \omega_5\}\}, \{\{\omega_1\}, \{\omega_2, \omega_4\}, \{\omega_3, \omega_5\}\}$ .
- The true state of the world would be  $\omega_2$ .

2. Romeo composes a letter to Juliet, and gives it to Tybalt to deliver to Juliet. While on the way, Tybalt peeks at the letter's contents. Tybalt gives Juliet the letter, and Juliet reads it immediately, in Tybalt's presence. Neither Romeo nor Juliet knows that Tybalt has read the letter. Answer the following questions about this story:
  - (a) Provide a knowledge space in which all the elements of the story above regarding the knowledge possessed by Romeo, Tybalt, and Juliet regarding the content of the letter hold true. Specify the state of the world in the model that corresponds to the situation described above.
  - (b) In the state of the world you specified above, does Romeo know that Juliet has read the letter? Justify.
  - (c) In the state of the world you specified above, does Tybalt know that Romeo knows that Juliet has read the letter? Justify.
  - (d) Construct a knowledge space in which, in addition to the particulars from the story presented above, the following also holds: "Tybalt does not know that Juliet does not know that Tybalt read the letter," and specify a state of the world in your model that corresponds to the situation.

## Solution

- (a)  $N = \{R, T, J\}$ . Consider 4 states of the world regarding knowledge of the contents of the letter possessed by the three players.
  - in  $\omega_1$  only Romeo knows the content of the letter;
  - in  $\omega_2$  only Romeo and Tybalt know the content of the letter;
  - in  $\omega_3$  only Romeo and Juliet know the content of the letter;
  - in  $\omega_4$  Romeo, Tybalt and Juliet know the content the letter.

Let  $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}$ . Player's partitions are then:

$$\mathcal{P}_R = \{\{\omega_1, \omega_2, \omega_3, \omega_4\}\},$$

$$\mathcal{P}_T = \{\{\omega_1\}, \{\omega_2\}, \{\omega_3\}, \{\omega_4\}\},$$

$$\mathcal{P}_J = \{\{\omega_1, \omega_2\}, \{\omega_3, \omega_4\}\}.$$

The true state of the world would be  $\omega_4$ .

- (b)  $K_R(\{\omega_3, \omega_4\}) = \emptyset$  so Romeo doesn't know that Juliet has read the letter in  $\omega_4$ .
- (c)  $K_T K_R(\{\omega_3, \omega_4\}) = \emptyset$ , so Tybalt does not know that Romeo knows that Juliet has read the letter in  $\omega_4$ . Also, since  $K_T((K_R(\{\omega_3, \omega_4\}))^c) = \Omega$ , so Tybalt knows that Romeo does not know that Juliet has read the letter.

- (d) In this new situation, we will need to add a new state of the world,  $\omega_4^*$  in which Romeo, Tybalt, and Juliet know the content of the letter, and Juliet knows that Tybalt read the letter. Thus, the set of states is  $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4, \omega_4^*\}$ , with the same set of players. The information partitions are:

$$\mathcal{P}_R = \{\{\omega_1, \omega_2, \omega_3, \omega_4, \omega_4^*\}\},$$

$$\mathcal{P}_T = \{\{\omega_1\}, \{\omega_2\}, \{\omega_3\}, \{\omega_4, \omega_4^*\}\},$$

$$\mathcal{P}_J = \{\{\omega_1, \omega_2\}, \{\omega_3, \omega_4\}, \{\omega_4^*\}\}.$$

The true state of the world remains  $\omega_4$ .